

Database Design :Authentication & Authorization

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System Users-

1. Principal,
2. Vice Principal
3. HOD
4. Faculty (Class & Subject Teachers)
5. Students
6. Accountant
7. Exam Controller
8. Training & Placement Cell
9. Physical Director
10. Admin
11. IT Admin
12. Visitors

1)Functional Requirements of Database

1. Register users
2. Store encrypted passwords
3. Authenticate users (username + password)
4. Assign roles to users
5. Assign permissions to roles
6. Restrict module access based on role
7. Maintain login history

2)Non-Functional Requirements of Database

1. High security(Achieved by RBAC, login_log)
2. Password hashing(Achieved by using hashing algorithms like bcrypt)
3. Scalable user handling(Separate tables for users, roles, and permissions)
4. Data integrity(Achieved by PK,FK,NOT NULL,UNIQUE CONSTRAINTS)

Core Entities of Database

1. users

Stores all registered system users with login credentials and account status.
Acts as the main authentication entity.

Field Name	Data Type	Constraints
user_id	INT	PRIMARY KEY, AUTO_INCREMENT
username	VARCHAR(50)	NOT NULL, UNIQUE
email	VARCHAR(100)	NOT NULL, UNIQUE
password_hash	VARCHAR(255)	NOT NULL
status	BOOLEAN	NOT NULL, DEFAULT TRUE
created_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

user_id	username	email	password_hash	status	created_at	updated_at
1	principal01	principal@school.com	hashed123	TRUE	2026-02-01 09:00	2026-02-01 09:00
2	teacher01	teacher@school.com	hashed456	TRUE	2026-02-02 10:00	2026-02-02 10:00
3	student01	student@school.com	hashed789	TRUE	2026-02-03 11:00	2026-02-03 11:00

2. roles

Stores predefined system roles (Principal, Faculty, Student, Admin, etc.).

Field Name	Data Type	Constraints
role_id	INT	PRIMARY KEY, AUTO_INCREMENT
role_name	VARCHAR(50)	NOT NULL, UNIQUE
description	TEXT	Optional (can be NULL)
created_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

role_id	role_name	description	created_at	updated_at
1	Principal	Head of school	2026-02-01 08:00	2026-02-01 08:00
2	Faculty	Teaching staff	2026-02-01 08:05	2026-02-01 08:05
3	Student	School student	2026-02-01 08:10	2026-02-01 08:10
4	Admin	System administrator	2026-02-01 08:15	2026-02-01 08:15

3. user_roles

links users and roles. Allows one user to have multiple roles.

Field Name	Data Type	Constraints
user_role_id	INT	PRIMARY KEY , AUTO_INCREMENT
user_id	INT	NOT NULL , FOREIGN KEY → users(user_id)
role_id	INT	NOT NULL , FOREIGN KEY → roles(role_id)
created_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

user_role_id	user_id	role_id	created_at	updated_at
1	1	1	2026-02-01 09:05	2026-02-01 09:05
2	1	4	2026-02-01 09:06	2026-02-01 09:06
3	2	2	2026-02-02 10:05	2026-02-02 10:05
4	3	3	2026-02-03 11:05	2026-02-03 11:05

4. permissions

Stores individual actions that can be performed (create, update, view, delete).

Field Name	Data Type	Constraints
permission_id	INT	PRIMARY KEY , AUTO_INCREMENT
permission_name	VARCHAR(100)	NOT NULL
action	VARCHAR(50)	NOT NULL
module_id	INT	NOT NULL, FOREIGN KEY → modules(module_id)
created_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

permission_id	permission_name	action	module_id	created_at	updated_at
1	View Dashboard	view	1	2026-02-01 07:30	2026-02-01 07:30
2	Mark Attendance	create	2	2026-02-01 07:35	2026-02-01 07:35
3	Edit Attendance	update	2	2026-02-01 07:40	2026-02-01 07:40
4	View Reports	view	3	2026-02-01 07:45	2026-02-01 07:45

5. role_permissions

Junction table that assigns permissions to roles. Defines what each role is allowed to do.

Field Name	Data Type	Constraints
role_permission_id	INT	PRIMARY KEY , AUTO_INCREMENT
role_id	INT	NOT NULL, FOREIGN KEY → roles(role_id)
permission_id	INT	NOT NULL, FOREIGN KEY → permissions(permission_id)
created_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

role_permission_id	role_id	permission_id	created_at	updated_at
1	1	1	2026-02-01 09:10	2026-02-01 09:10
2	1	4	2026-02-01 09:11	2026-02-01 09:11
3	2	2	2026-02-02 10:10	2026-02-02 10:10
4	2	3	2026-02-02 10:11	2026-02-02 10:11
5	3	1	2026-02-03 11:10	2026-02-03 11:10

6. modules

Stores different system modules (Dashboard, etc.).
Permissions are grouped under modules.

Field Name	Data Type	Constraints
module_id	INT	PRIMARY KEY , AUTO_INCREMENT
module_name	VARCHAR(100)	NOT NULL , UNIQUE
description	VARCHAR(255)	Optional (can be NULL)
created_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

module_id	module_name	description	created_at	updated_at
1	Dashboard	Main system page	2026-02-01 07:00	2026-02-01 07:00
2	Attendance	Manage attendance	2026-02-01 07:10	2026-02-01 07:10
3	Reports	View system reports	2026-02-01 07:20	2026-02-01 07:20

7.refresh_tokens

Field Name	Data Type	Constraints
refresh_token_id	INT	PRIMARY KEY , AUTO_INCREMENT
user_id	INT	NOT NULL , FOREIGN KEY → users(user_id)
token	VARCHAR(255)	NOT NULL , UNIQUE
expiry_date	TIMESTAMP	NOT NULL
created_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

refresh_token_id	user_id	token	expiry_date	created_at	updated_at
1	1	tokenABC	2026-03-01 00:00	2026-02-10 09:00	2026-02-10 09:00
2	1	tokenXYZ	2026-03-01 00:00	2026-02-10 09:05	2026-02-10 09:05
3	2	tokenLMN	2026-03-01 00:00	2026-02-11 10:00	2026-02-11 10:00

8.login_log

Tracks user login activity including login time, IP address, and device information.
Used for auditing and security monitoring.

Field Name	Data Type	Constraints
login_log_id	INT	PRIMARY KEY , AUTO_INCREMENT
user_id	INT	NOT NULL , FOREIGN KEY → users(user_id)
ip_address	VARCHAR(45)	NOT NULL
device_info	VARCHAR(255)	Optional (can be NULL)
status	VARCHAR(50)	NOT NULL
login_time	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP
created_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NOT NULL , DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Example-

login_log_id	user_id	ip_address	device_info	status	login_time	created_at	updated_at
1	1	192.168.1.10	Chrome PC	Success	2026-02-10 09:00	2026-02-10 09:00	2026-02-10 09:00
2	2	192.168.1.15	Android	Success	2026-02-11 10:00	2026-02-11 10:00	2026-02-11 10:00
3	3	192.168.1.20	Chrome PC	Failed	2026-02-12 11:00	2026-02-12 11:00	2026-02-12 11:00

Relationships

1)Users → User_Roles (1:M)

One user can have multiple roles in the system.

2)Roles → User_Roles (1:M)

One role can be assigned to many users.

This creates a many-to-many relationship between users and roles.

3)Roles → Role_Permissions (1:M)

One role can have multiple permissions.

Each record defines what actions that role is allowed to perform.

4)Permissions → Role_Permissions (1:M)

One permission can be given to many roles.

This creates a many-to-many relationship between roles and permissions.

5)Modules → Permissions (1:M)

One module contains multiple permissions (like create, edit, delete).

Each permission belongs to only one module.

6)Users → Refresh_Tokens (1:M)

One user can have multiple refresh tokens (for multiple devices).

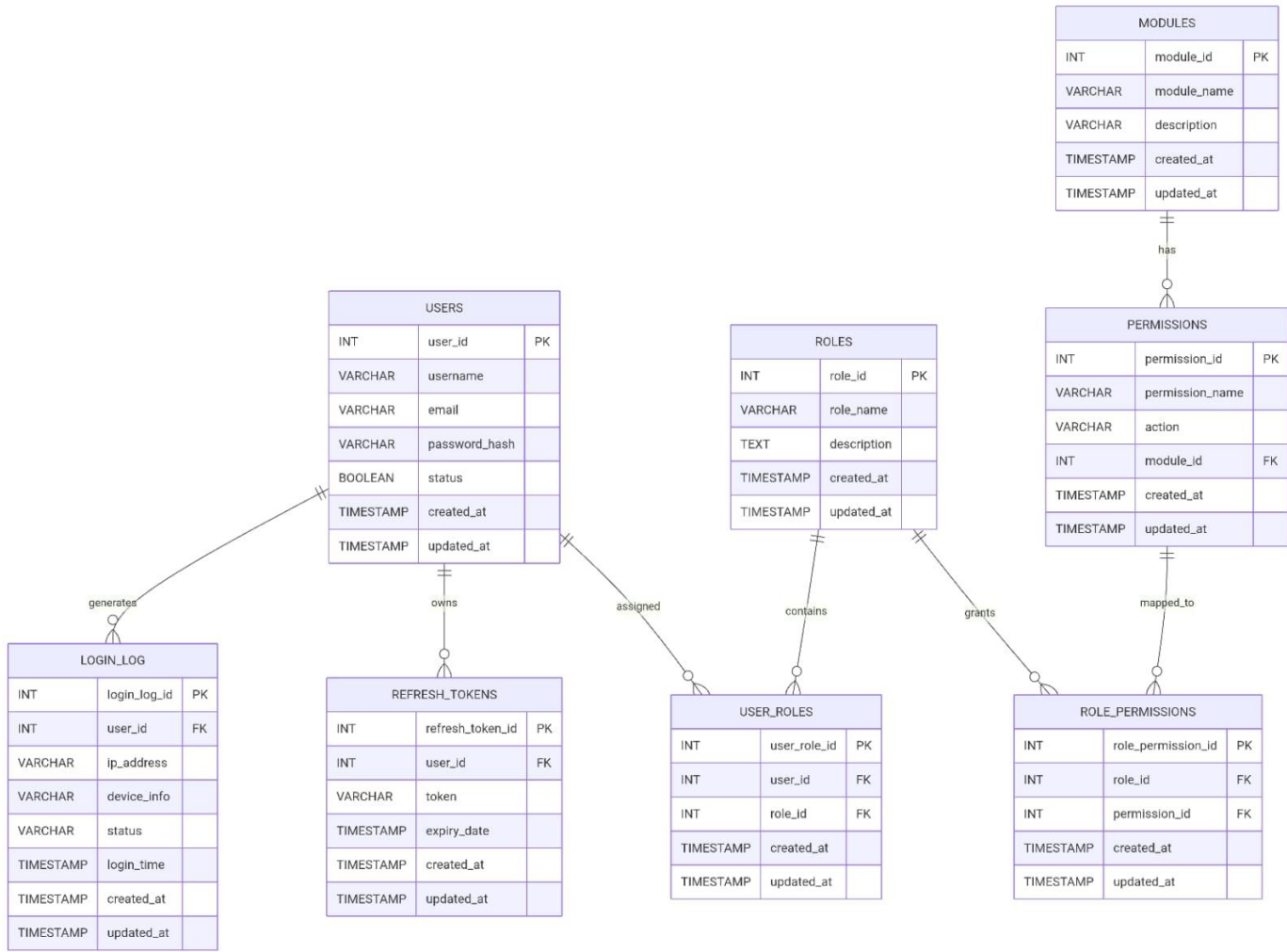
Each token belongs to only one user.

7)Users → Login_Log (1:M)

One user can have many login records.

Each login log entry belongs to one user.

Entity Relationship Diagram (ERD)



Login Flow

